

Classical Electrodynamics (PHYS 5331) Final Exam

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1 Question 1

Question : Solve Laplace equation in cylindrical coordinates

Question 1 Answer: The definition of the Laplace Equation

$$\nabla^2\Phi = 0 \quad (1)$$

The definition of ∇ in cylindrical coordinates

$$\nabla = \hat{e}_1 \frac{\partial}{\partial \rho} + \hat{e}_2 \frac{1}{\rho} \frac{\partial}{\partial \phi} + \hat{e}_3 \frac{\partial}{\partial z} \quad (2)$$

Combining the two

$$\nabla^2\Phi = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \Phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \Phi}{\partial \theta^2} + \frac{\partial^2 \Phi}{\partial z^2} = 0 \quad (3)$$

Expand the r term

$$\frac{1}{r} \frac{\partial \Phi}{\partial r} + \frac{\partial^2 \Phi}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2 \Phi}{\partial \theta^2} + \frac{\partial^2 \Phi}{\partial z^2} = 0 \quad (4)$$

Using separation of variables, we assume that the solution is of the form

$$\Phi(r, \theta, z) = R(r)P(\theta)Z(z) \quad (5)$$

Substitute equation 5 into equation 4

$$\frac{1}{r} \frac{\partial(RPZ)}{\partial r} + \frac{\partial^2(RPZ)}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2(RPZ)}{\partial \theta^2} + \frac{\partial^2(RPZ)}{\partial z^2} = 0 \quad (6)$$

Bring variables that are not derivable out of derivative, and change $\partial \rightarrow d$

$$\frac{PZ}{r} \frac{dR}{dr} + PZ \frac{d^2 R}{dr^2} + \frac{RZ}{r^2} \frac{d^2 P}{d\theta^2} + RP \frac{d^2 Z}{dz^2} = 0 \quad (7)$$

To isolate the variables, divide both sides by $\Phi = RPZ$

$$\frac{1}{Rr} \frac{dR}{dr} + \frac{1}{R} \frac{d^2 R}{dr^2} + \frac{1}{Pr^2} \frac{d^2 P}{d\theta^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = 0 \quad (8)$$

Isolate Z and equate it to a constant λ^2

$$\frac{1}{Rr} \frac{dR}{dr} + \frac{1}{R} \frac{d^2 R}{dr^2} + \frac{1}{Pr^2} \frac{d^2 P}{d\theta^2} = -\frac{1}{Z} \frac{d^2 Z}{dz^2} = \lambda \quad (9)$$

To further isolate variables, multiply both sides by r^2

$$\frac{r}{R} \frac{dR}{dr} + \frac{r^2}{R} \frac{d^2 R}{dr^2} + \frac{1}{P} \frac{d^2 P}{d\theta^2} = \lambda r^2 \quad (10)$$

Isolate P and equate it to a constant κ^2

$$\frac{r}{R} \frac{dR}{dr} + \frac{r^2}{R} \frac{d^2 R}{dr^2} - \lambda r^2 = -\frac{1}{P} \frac{d^2 P}{d\theta^2} = \kappa \quad (11)$$

Place λ in a differential forms

$$-\frac{1}{Z} \frac{d^2 Z}{dz^2} = \lambda^2 \quad (12)$$

Put all variable to one side

$$\frac{d^2 Z}{dz^2} = Z\lambda^2 \quad (13)$$

Place κ in a differential forms

$$-\frac{1}{P} \frac{d^2 P}{d\theta^2} = \kappa \quad (14)$$

Put all variable to one side

$$\frac{d^2 P}{d\theta^2} = -\kappa P \quad (15)$$

To find the general case we will consider four cases, $\kappa = 0$ and $\lambda = 0$, $\kappa = 0$ and $\lambda \neq 0$, $\kappa \neq 0$ and $\lambda = 0$, and $\kappa \neq 0$ and $\lambda \neq 0$. Once these solutions are determined we will combine them to find the general solution.

Case 1 ($\kappa = 0$ and $\lambda = 0$):

$$\begin{aligned} R(r) &= A_{0,0} + B_{0,0} \ln(r) \\ P(\phi) &= C_{0,0} + D_{0,0} \phi \\ Z(z) &= F_{0,0} + G_{0,0} z \end{aligned} \quad (16)$$

$$\Phi_1 = (A_{0,0} + B_{0,0} \ln(r))(C_{0,0} + D_{0,0} \phi)(F_{0,0} + G_{0,0} z) \quad (17)$$

Case 2 ($\kappa \neq 0$ and $\lambda = 0$):

$$\begin{aligned} R(r) &= A_{\kappa,0}r^\kappa + B_{\kappa,0}r^{-\kappa} \\ P(\phi) &= C_{\kappa,0}e^{i\kappa\phi} + D_{\kappa,0}e^{-i\kappa\phi} \\ Z(z) &= F_{\kappa,0} + G_{\kappa,0}z \end{aligned} \quad (18)$$

$$\Phi_2 = (A_{\kappa,0}r^\kappa + B_{\kappa,0}r^{-\kappa})(C_{\kappa,0}e^{i\kappa\phi} + D_{\kappa,0}e^{-i\kappa\phi})(F_{\kappa,0} + G_{\kappa,0}z) \quad (19)$$

For the following cases where $\lambda \neq 0$, the solution of R must be solved using a series solution Case 3 ($\kappa = 0$ and $\lambda \neq 0$):

$$\begin{aligned} P(\phi) &= C_{0,\lambda} + D_{0,\lambda}\phi \\ Z(z) &= F_{0,\lambda}e^{\lambda z} + G_{0,\lambda}e^{-\lambda z} \end{aligned} \quad (20)$$

Case 4 ($\kappa \neq 0$ and $\lambda \neq 0$):

$$\begin{aligned} P(\phi) &= C_{\kappa,\lambda}e^{\kappa\phi} + D_{\kappa,\lambda}e^{-\kappa\phi} \\ Z(z) &= F_{\kappa,\lambda}e^{\lambda z} + G_{\kappa,\lambda}e^{-\lambda z} \end{aligned} \quad (21)$$

Put equation 11 in more intuitive form

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dR}{dr} \right) + \left(\lambda^2 - \frac{\kappa^2}{r^2} \right) R = 0 \quad (22)$$

Substitute $x = \lambda r$

$$\frac{1}{x} \frac{d}{dx} \left(x \frac{dR}{dx} \right) + \left(1 - \frac{\kappa^2}{x^2} \right) R = 0 \quad (23)$$

Guess, or "Try", the solution

$$R = \sum_{j=0}^{\infty} a_j x^{j+\alpha} \quad (24)$$

Substitute and derive

$$\sum_{j=0}^{\infty} a_j x^{j+\alpha-2} ((j+\alpha) - \kappa^2) + \sum_{j=0}^{\infty} a_j x^{j+\alpha} = 0 \quad (25)$$

Simplify

$$a_0 x^{\alpha-2} (\alpha^2 - \kappa^2) + a_1 x^{\alpha-1} ((1+\alpha)^2 - \kappa^2) + \sum_{j=0}^{\infty} x^{j+\alpha} (a_{j+2} ((j+2+\alpha)^2 - \kappa^2) a_j) = 0 \quad (26)$$

Every power of x is independent, thus the coefficient of every power vanishes.

$$\begin{aligned} a_0 (\alpha^2 - \kappa^2) &= 0 \\ a_1 ((1+\alpha)^2 - \kappa^2) &= 0 \\ a_{j+2} &= - \frac{a_j}{((j+2+\alpha)^2 - \kappa^2)} \end{aligned} \quad (27)$$

This matches the first Bessel function (J)

$$R = J_\kappa = \sum_{j=1}^{\infty} a_{2(j-1)} x^{2(j-1)+\kappa} \quad (28)$$

Where

$$a_{2j} = -\frac{a_{2(j-1)}}{4j(j+\kappa)} \quad (29)$$

The other solution is the second Bessel function (N)

$$N_\kappa = \frac{J_\kappa \cos(\kappa\pi) - J_{-\kappa}}{\sin(\kappa\pi)} \quad (30)$$

Combine these to get the general equation

$$R = A_{\kappa,\lambda} J_\kappa + B A_{\kappa,\lambda} N_\kappa \quad (31)$$

Combine all four cases

$$\begin{aligned} \Phi(r, \phi, z) = & (A_{0,0} + B_{0,0} \ln(r))(C_{0,0} + D_{0,0} \phi)(F_{0,0} + G_{0,0} z) \\ & + \sum_{\kappa \neq 0} (A_{\kappa,0} r^\kappa + B_{\kappa,0} r^{-\kappa})(C_{\kappa,0} e^{i\kappa\phi} + D_{\kappa,0} e^{-i\kappa\phi})(F_{\kappa,0} + G_{\kappa,0} z) \\ & + \sum_{\lambda \neq 0} (A_{0,\lambda} J_0 + B_{0,\lambda} N_0)(C_{0,\lambda} + D_{0,\lambda} \phi)(F_{0,\lambda} e^{\lambda z} + G_{0,\lambda} e^{-\lambda z}) \quad (32) \\ & + \sum_{\kappa \neq 0} \sum_{\lambda \neq 0} (A_{\kappa,\lambda} J_\kappa + B_{\kappa,\lambda} N_\kappa)(C_{\kappa,\lambda} e^{\kappa\phi} + D_{\kappa,\lambda} e^{-\kappa\phi})(F_{\kappa,\lambda} e^{\lambda z} + G_{\kappa,\lambda} e^{-\lambda z}) \end{aligned}$$

2 Question 2

Question : Solve problem 3.6 from the textbook for $a = 2nm$ for a charge equal to charge of an electron

Problem 3.6 Question

Two point charges q and $-q$ are located on the z axis at $z = +a$ and $z = -a$, respectively.

- Find the electrostatic potential as an expansion in spherical harmonics and powers of r for both $r > a$ and $r < a$
- Keeping the product $qa \equiv \frac{\rho}{2}$ constant, take the limit of $a \rightarrow 0$ and find the potential for $r \neq 0$. This is by definition a dipole along the z axis and its potential.
- Suppose now that the dipole of part b is surrounded by a grounded spherical shell of radius b concentric with the origin. By linear superposition find the potential everywhere inside the shell

Question 2, Section a) Answer: Integrate equation 114 from Appendix 1 (section 6.1). The charges are located on the z axis so two of the axis have no component.

$$\Phi = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{|\vec{r} - a\hat{k}|} - \frac{q}{|\vec{r} + a\hat{k}|} \right) \quad (33)$$

Use the addition theorem where $r_<$ is smaller of (r, a) , and $r_>$ is larger of (r, a)

$$\frac{1}{|\vec{r} - \vec{r}'|} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_<^l}{r_>^{l+1}} Y_{lm}(\theta', \phi') Y_{lm}(\theta, \phi) \quad (34)$$

Apply addition theorem, using $\phi' = 0$ for both cases,

$$\text{and } \theta' = \begin{cases} 0 & \text{for } r - a\hat{k} \\ \pi & \text{for } r + a\hat{k} \end{cases}$$

because the charges are on the same axis.

$$\Phi = \frac{q}{4\pi\epsilon_0} \left(4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_<^l}{r_>^{l+1}} Y_{lm}^*(0, 0) Y_{lm}(\theta, \phi) - 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_<^l}{r_>^{l+1}} Y_{lm}^*(\pi, 0) Y_{lm}(\theta, \phi) \right) \quad (35)$$

Simplify

$$\Phi = \frac{q}{\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_<^l}{r_>^{l+1}} Y_{lm}(\theta, \phi) (Y_{lm}^*(0, 0) - Y_{lm}^*(\pi, 0)) \quad (36)$$

$$\Phi = \begin{cases} \frac{q}{\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_l^l}{a^{l+1}} Y_{lm}(\theta, \phi) (Y_{lm}^*(0, 0) - Y_{lm}^*(\pi, 0)) & \text{when } r < a \\ \frac{q}{\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{a^l}{r^{l+1}} Y_{lm}(\theta, \phi) (Y_{lm}^*(0, 0) - Y_{lm}^*(\pi, 0)) & \text{when } r > a \end{cases} \quad (37)$$

Question 2, Section Answer b): Simplify equation 36 by applying azimuthally symmetric condition, all terms other than $m = 0$ vanish.

$$\Phi = \frac{q}{\epsilon_0} \sum_{l=0}^{\infty} \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{lm}(\theta, \phi) (Y_{lm}^*(0, 0) - Y_{lm}^*(\pi, 0)) \quad (38)$$

Convert Spherical harmonics into Legendre polynomial

$$\Phi = \frac{q}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{r_{<}^l}{r_{>}^{l+1}} P_l(\cos(\theta)) (P_l(1) - P_l(-1)) \quad (39)$$

Evaluate Legendre polynomial

$$\Phi = \frac{q}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{r_{<}^l}{r_{>}^{l+1}} P_l(\cos(\theta)) (1 - (-1)^{-1}) \quad (40)$$

Simplify and alter iteration

$$\Phi = \frac{2q}{4\pi\epsilon_0} \sum_{l=0, \text{odd}}^{\infty} \frac{r_{<}^l}{r_{>}^{l+1}} P_l(\cos(\theta)) \quad (41)$$

Take the limit of $\alpha \rightarrow 0$ and apply $r \neq 0$ and $r > a$

$$\Phi = \frac{2q}{4\pi\epsilon_0} \sum_{l=0, \text{odd}}^{\infty} \frac{a^l}{r^{l+1}} P_l(\cos(\theta)) \quad (42)$$

$$\Phi = \frac{p}{4\pi\epsilon_0} \sum_{l=0, \text{odd}}^{\infty} \frac{a^{l-1}}{r^{l+1}} P_l(\cos(\theta)) \quad (43)$$

$$\Phi = \frac{p}{4\pi\epsilon_0} \left(\frac{1}{r^2} P_1(\cos(\theta)) + \frac{a^2}{r^4} P_3(\cos(\theta)) + \frac{a^4}{r^6} P_5(\cos(\theta)) + \dots \right) \quad (44)$$

Take limit $\alpha \rightarrow 0$

$$\Phi = \frac{p}{4\pi\epsilon_0} \frac{1}{r^2} \cos(\theta) \quad (45)$$

Apply condition that $a = 2 \text{ [nm]} = 2 \times 10^{-9} \text{ [m]}$ to given constant and $q = a$

$$p = 2qa = 2 - e(2\text{nm}) = 6.41 \times 10^{-28} \text{ [C*m]} \quad (46)$$

$$\Phi = \frac{6.41 \times 10^{-28} \text{ [C*m]}}{4\pi\epsilon_0} \frac{1}{r^2} \cos(\theta) \quad (47)$$

Question 2, Section Answer c): Then if dipole is surrounded by a grounded sphere shell sharing an origin, there will be additional potential from the sphere. Using equation 45, add potential from sphere. The potential from the surface of a sphere can be derived from the Laplace equation in spherical coordinates (From

appendix 2, section 6.2, specifically equation 158

$$\Phi(r, \theta, \phi) = \sum_{l=0}^{\infty} (A_l r^l + B_l r^{-l-1}) P_l(\cos(\theta))$$

Where $B_l = 0$ because the region of valid solution includes the origin.

$$\Phi(r, \theta, \phi) = \sum_{l=0}^{\infty} A_l r^l P_l(\cos(\theta)) \quad (48)$$

Add potential from sphere into equation 45

$$\Phi = \frac{p}{4\pi\epsilon_0} \frac{1}{r^2} \cos(\theta) + \sum_{l=0}^{\infty} A_l r^l P_l(\cos(\theta)) \quad (49)$$

Apply boundary condition; potential is zero at surface of sphere. Therefore, $\Phi = 0$ if $r = b$ where b is the radius of the sphere.

$$0 = \frac{p}{4\pi\epsilon_0} \frac{1}{b^2} \cos(\theta) + \sum_{l=0}^{\infty} A_l b^l P_l(\cos(\theta)) \quad (50)$$

Only $l = 1$ will be nonzero because of orthogonality

$$0 = \frac{p}{4\pi\epsilon_0} \frac{1}{b^2} \cos(\theta) + A_l b \cos(\theta) \quad (51)$$

Isolate A_l

$$A_l = -\frac{p}{4\pi\epsilon_0} \frac{1}{b^3} \quad (52)$$

Plug A_l into equation 49

$$\Phi = \frac{p}{4\pi\epsilon_0} \frac{1}{r^2} \cos(\theta) + \sum_{l=0}^{\infty} -\frac{p}{4\pi\epsilon_0} \frac{1}{b^3} r^l P_l(\cos(\theta)) \quad (53)$$

Again, Only $l = 1$ will be nonzero because of orthogonality

$$\Phi = \frac{p \cos(\theta)}{4\pi\epsilon_0 b^2} \left(\left(\frac{b}{r} \right)^2 - \frac{r}{b} \right) \quad (54)$$

Similarly apply the condition from the given value

$$\Phi = \frac{6.41 \times 10^{-28} [\text{C} \cdot \text{m}] \cos(\theta)}{4\pi\epsilon_0 b^2} \left(\left(\frac{b}{r} \right)^2 - \frac{r}{b} \right) \quad (55)$$

3 Question 3

Question : Solve Problem 4.13 from the textbook

Problem 4.13 question

Two long, coaxial, cylindrical conducting surfaces of radii a and b are lowered vertically into a liquid dielectric. If the liquid rises an average height h between the electrodes when a potential difference V is established between them, show that the susceptibility of the liquid is

$$\chi_e = \frac{(b^2 - a^2)\rho gh \ln(\frac{b}{a})}{\epsilon_0 V^2}$$

where ρ is the density of the liquid, g is the acceleration due to gravity, and the susceptibility of air is neglected

Question 3 Answer: The general solution to the laplace equations in cylindrical coordinates (See section 1 for formulation)

$$\begin{aligned} \Phi(r, \phi, z) = & (A_{0,0} + B_{0,0} \ln(r))(C_{0,0} + D_{0,0} \phi)(F_{0,0} + G_{0,0} z) \\ & + \sum_{\kappa \neq 0} (A_{\kappa,0} r^\kappa + B_{\kappa,0} r^{-\kappa})(C_{\kappa,0} e^{i\kappa\phi} + D_{\kappa,0} e^{-i\kappa\phi})(F_{\kappa,0} + G_{\kappa,0} z) \\ & + \sum_{\lambda \neq 0} (A_{0,\lambda} J_0 + B_{0,\lambda} N_0)(C_{0,\lambda} + D_{0,\lambda} \phi)(F_{0,\lambda} e^{\lambda z} + G_{0,\lambda} e^{-\lambda z}) \\ & + \sum_{\kappa \neq 0} \sum_{\lambda \neq 0} (A_{\kappa,\lambda} J_\kappa + B_{\kappa,\lambda} N_\kappa)(C_{\kappa,\lambda} e^{\kappa\phi} + D_{\kappa,\lambda} e^{-\kappa\phi})(F_{\kappa,\lambda} e^{\lambda z} + G_{\kappa,\lambda} e^{-\lambda z}) \end{aligned}$$

Azimuthal symmetry gets rid of most terms, and we will consider $r = \rho$, $A_{0,0} = a_0$, and $B_{0,0} = b_0$.

$$\Phi(\rho, \phi) = a_0 + b_0 \ln(\rho) \quad (56)$$

Apply boundary condition at inner cylinder $\Phi(\rho = a) = 0$

$$0 = a_0 + b_0 \ln(a) \quad (57)$$

Isolate a_0

$$a_0 = -b_0 \ln(a) \quad (58)$$

Substitute back into equation

$$\Phi(\rho, \phi) = b_0 \ln\left(\frac{\rho}{a}\right) \quad (59)$$

Apply boundary condition at outer cylinder $\Phi(\rho = b) = V$

$$V = a_0 + b_0 \ln(b) \quad (60)$$

Isolate b_0

$$b_0 = \frac{V}{\ln(\frac{b}{a})} \quad (61)$$

Substitute back into equation

$$\Phi(\rho, \phi) = V \frac{\ln(\frac{\rho}{a})}{\ln(\frac{b}{a})} \quad (62)$$

Using definition of electric field, substitute

$$\vec{E} = -\rho V \frac{1}{\rho \ln(\frac{b}{a})} \quad (63)$$

Using the conservation of energy

$$\Delta W = W_f - W_i \quad (64)$$

Use definition of potential energy

$$\Delta W = \frac{1}{2} \int E_f \cdot D_f d^3x - \frac{1}{2} \int E_i \cdot D_i d^3x \quad (65)$$

Evaluate dot product, remove non-integrateable variables from within the integral, and substitute situational electric fields

$$\Delta W = \left(\frac{1}{2} (L - h) \epsilon_0 \int E_{air}^2 d^2x + \frac{1}{2} \epsilon_0 (1 + \chi_e) h \int E_{liqu}^2 d^2x \right) - \left(\frac{1}{2} L \epsilon_0 \int E_{air}^2 d^2x \right) \quad (66)$$

Simplify

$$\Delta W = -\frac{1}{2} h \epsilon_0 \int E_{air}^2 d^2x + \frac{1}{2} \epsilon_0 (1 + \chi_e) h \int E_{liqu}^2 d^2x \quad (67)$$

Assume $E_{air} = E_{liqu}$

$$\Delta W = \frac{h \epsilon_0 \chi_e}{2} \int E^2 d^2x \quad (68)$$

$$\Delta W = h \epsilon_0 \chi_e \pi \int_a^b E^2 \rho d\rho \quad (69)$$

Substitute equation

$$\Delta W = h \epsilon_0 \chi_e \pi \int_a^b \left(-\hat{\rho}^2 V^2 \frac{1}{\rho^2 \ln^2(\frac{b}{a})} \right) \rho d\rho \quad (70)$$

Evaluate integral

$$\Delta W = \frac{\pi h \epsilon_0 \chi_e V^2}{\ln(\frac{b}{a})} \quad (71)$$

We can view the potential energy gain as being held in the particles

$$\Delta W = mgh \quad (72)$$

Substitute definition of mass $m = \rho h A$

$$\Delta W = \rho h A g h \quad (73)$$

Substitute geometric surface area of the two cylinders

$$\Delta W = \rho \pi (b^2 - a^2) g h^2 \quad (74)$$

Equate equation

$$\frac{\pi h \epsilon_0 \chi_e V^2}{\ln(\frac{b}{a})} = \rho \pi (b^2 - a^2) g h^2 \quad (75)$$

Isolate χ_e

$$\chi_e = \frac{(b^2 - a^2) \rho g h \ln(\frac{b}{a})}{\epsilon_0 V^2} \quad (76)$$

4 Question 4

Question : Solve problem 5.22 from the textbook and find the magnetization due to an applied force of 2.5 [N] in the right units

Problem 5.22 question

Show that in general a long, straight bar of uniform cross-sectional area A with uniform lengthwise magnetization M , when placed with its flat end against an infinitely permeable flat surface, adheres with a force given approximately by

$$F \simeq \frac{\mu_0}{2} AM^2$$

Relate your discussion to the electrostatic considerations in Section 1.11. Question 4 Answer:

This bar has an arbitrary cross sectional shape extending into the $\hat{z}$ direction with the infinitely permeable flat surface, or perfect mirror, in the $\hat{x} - \hat{y}$ plane.

$$\sigma_M = \vec{n} \cdot \vec{M} \quad (77)$$

Assume the bar is extending into the $\hat{z}$ direction

$$\sigma_M = -\hat{z} \cdot (M\hat{z}) \quad (78)$$

At the bottom face of the bar $z = 0$

$$\sigma_M(z = 0) = -M \quad (79)$$

The infinitely permeable flat surface, or perfect mirror, is in the $\hat{x} - \hat{y}$ plane so we can place an image bar on the opposite side, with an opposite magnetization

$$\sigma'_M(z = 0) = M \quad (80)$$

Because both bars are "long" we can assume that the fields just past the ends are equivalent to the bars surface

$$\vec{B} = \mu_0 \vec{H} + \mu_0 \vec{M} \quad (81)$$

Inside an infinitely long bar $H = 0$

$$B = \mu_0 M \quad (82)$$

By definition, the total force over a surface charge is

$$\vec{F} = \int_S \sigma_M \vec{B} da \quad (83)$$

Substitute

$$\vec{F} = \int_S (-M)(\mu_0 M \hat{z}) da \quad (84)$$

Integrate

$$\vec{F} = -\hat{z} \mu_0 AM^2 \quad (85)$$

Remove influence from image bar

$$\vec{F} = -\frac{\vec{z}\mu_0 AM^2}{2} \quad (86)$$

5 Question 5

Question : Derive the Poynting vector and describe its physical significance

Question 5 Answer: A single particle (q) travels with the velocity ($\vec{v}$) through an electric ($\vec{E}$) and magnetic ($\vec{B}$) external field.

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) \quad (87)$$

By definition, Ampere-Maxwell's law states

$$\vec{J} = \frac{c}{4\pi} \nabla \times \vec{H} - \frac{1}{4\pi} \frac{\partial \vec{D}}{\partial t} \quad (88)$$

By definition the work done by charges in a volume

$$W = \frac{1}{4\pi} \int_V \left(c\vec{E} \cdot (\nabla \times \vec{H}) - \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \right) dV = \int_V \vec{J} \cdot \vec{E} dV \quad (89)$$

Using the Vector Identity

$$\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$$

Rewrite vector identity to match equation

$$\vec{A} \cdot (\nabla \times \vec{B}) = -(\nabla \cdot (\vec{A} \times \vec{B})) + \vec{B} \cdot (\nabla \times \vec{A}) \quad (90)$$

Substitute

$$\int_V \vec{J} \cdot \vec{E} dV = \frac{1}{4\pi} \int_V \left(-c\nabla \cdot (\vec{E} \times \vec{H}) + c\vec{H} \cdot (\nabla \times \vec{E}) - \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \right) dV \quad (91)$$

Using Faradays law

$$\nabla \times \vec{E} = -\frac{1}{c} \frac{\partial \vec{B}}{\partial t}$$

Substitute

$$\int_V \vec{J} \cdot \vec{E} dV = \frac{1}{4\pi} \int_V \left(-c\nabla \cdot (\vec{E} \times \vec{H}) + c\vec{H} \cdot \left(-\frac{1}{c} \frac{\partial \vec{B}}{\partial t} \right) - \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \right) dV \quad (92)$$

A portion of this equation relates to the total Energy density ($\vec{U}$) where the dielectric ($\vec{D} = \epsilon \vec{E}$) and the permeability ($\vec{H} = \frac{1}{\mu} \vec{B}$)

$$\vec{U} = \frac{1}{8\pi} (\vec{E} \cdot \vec{D} + \vec{B} \cdot \vec{H}) \quad (93)$$

Derivative with respect to time

$$\frac{\partial \vec{U}}{\partial t} = \frac{1}{4\pi} \left(\epsilon \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} + \frac{1}{\mu} \vec{B} \cdot \frac{\partial \vec{B}}{\partial t} \right) \quad (94)$$

Using the relation of Dielectric and permeability

$$\frac{\partial \vec{U}}{\partial t} = \frac{1}{4\pi} \left(\vec{E} \cdot \frac{\partial \vec{D}}{\partial t} + \vec{B} \cdot \frac{\partial \vec{H}}{\partial t} \right) \quad (95)$$

Substitute equation 95 into equation 92

$$\int_V \vec{J} \cdot \vec{E} dV = \int_V \left(\frac{c}{4\pi} \nabla \cdot (\vec{E} \times \vec{H}) + \frac{\partial \vec{U}}{\partial t} \right) dV \quad (96)$$

For convenience, we will define the pointing vector ($\vec{P}$)

$$P = \frac{c}{4\pi} (\vec{E} \times \vec{H}) \quad (97)$$

Substitute

$$\int_V \vec{J} \cdot \vec{E} dV = \int_V \left(\nabla \cdot \vec{P} + \frac{\partial \vec{U}}{\partial t} \right) dV \quad (98)$$

Since the volume element is arbitrary

$$-\vec{E} \cdot \vec{J} = \frac{\partial \vec{U}}{\partial t} + \nabla \cdot \vec{P} \quad (99)$$

The work done by external electric field

$$W_E = q(\vec{v} \cdot \vec{E}) \quad (100)$$

The work done by the external magnetic field

$$W_M = q(\vec{v} \times \vec{B}) = 0 \quad (101)$$

6 Appendix

6.1 Formulation of Electric Potential

Two Electric point charges exert a force on each other.

$$\vec{F} = q\vec{E} = kq_1q_2 \frac{x_1 - x_2}{|x_1 - x_2|^3} \quad (102)$$

Electric field at point x due to a point charge

$$\vec{E} = kq_1 \frac{x - x_1}{|x - x_1|^3} \quad (103)$$

In electrostatic units (esu) $k = 1$, In SI:

$$k = \frac{1}{4\pi\epsilon_0} = 10^{-7} c^2 \quad (104)$$

Since Electrodynamics is a macroscopic theory, we will consider the "point" to be many charged sources. Experiment shows that the fields add linearly.

$$\vec{E}(x) = k \sum_i q_i \frac{x - x_i}{|x - x_i|^3} \quad (105)$$

Consider the many charged sources to be "small" (Replace q with ρ) and "numerous" (Replace Sum with Integral) at point x' . Coulomb's law in terms of the field for a charge density. Replace singular x dimension with 3D x, y, z dimension

$$\vec{E}(x) = k \int \rho_q(x') \frac{x - x'}{|x - x'|^3} d^3 x' \quad (106)$$

A vector field can be specified completely if its divergence and curl are given everywhere in space, except up to the gradient of scalar function that satisfies the Laplace equation.

Therefore we need to specify the curl of $\vec{E}$

Starting at Coulomb's generalized integral form

$$\vec{E}(x) = k \int \rho_q(x') \frac{x - x'}{|x - x'|^3} d^3 x'$$

Using Property of Nabla (∇)

$$-\nabla \left(\frac{1}{|x - x'|} \right) = \frac{x - x'}{|x - x'|^3} \quad (107)$$

Substitute equation 104

$$\vec{E}(x) = k \int \rho_q(x') -\nabla \left(\frac{1}{|x - x'|} \right) d^3 x' \quad (108)$$

Nabla (∇) is constant; therefore,

$$\vec{E}(x) = -k \nabla \int \rho_q(x') \left(\frac{1}{|x - x'|} \right) d^3 x' \quad (109)$$

Cross nabla (∇) on both sides of equation and substitute equation

$$\nabla \times E(\vec{x}) = \nabla \times k - k \nabla \int \rho_q(x') \left(\frac{1}{|x - x'|} \right) d^3x' \quad (110)$$

Therefore,

$$\nabla \times \vec{E} = 0 \quad (111)$$

By manipulating the equation

$$E(\vec{x}) = \nabla \int \left(\frac{k \rho_q(x')}{|x - x'|} \right) d^3x' \quad (112)$$

We can now define a portion of the equation as the scalar potential (Φ)

$$E(\vec{x}) = \nabla \Phi \quad (113)$$

Therefore the scalar potential is defined as

$$\Phi = \int \left(\frac{k \rho_q(x')}{|x - x'|} \right) d^3x' \quad (114)$$

6.2 Laplace Equation in Spherical Coordinates

The definition of the Laplace equation

$$\nabla^2 \Phi = 0 \quad (115)$$

The definition of ∇ in spherical coordinates

$$\nabla = \hat{e}_1 \frac{\partial}{\partial r} + \hat{e}_2 \frac{1}{\theta} \frac{\partial}{\partial \theta} + \hat{e}_3 \frac{1}{r \sin(\theta)} \frac{\partial}{\partial \phi} \quad (116)$$

Where r is the radial distance from the origin, θ is the polar angle with the z-axis, ϕ is the azimuthal angle in the x-y plan relative to the x-axis

Combine the two

$$\frac{1}{r} \frac{\partial^2}{\partial r^2} (r \Phi) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial \Phi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2 \Phi}{\partial \phi^2} = 0 \quad (117)$$

Use method of separation of variables by trying a solution of the form

$$\Phi(r, \theta, \phi) = \frac{R(r)}{r} P(\theta) Q(\phi) \quad (118)$$

Substitute equation 118 into equation 117

$$\frac{1}{r} \frac{\partial^2}{\partial r^2} \left(r \frac{R}{r} P Q \right) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial}{\partial \theta} \left(\frac{R}{r} P Q \right) \right) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2}{\partial \phi^2} \left(\frac{R}{r} P Q \right) = 0 \quad (119)$$

Multiply by $\frac{r^3 \sin^2(\theta)}{RPQ}$

$$\frac{r^2 \sin^2(\theta)}{R} \frac{d^2 R}{dr^2} + \frac{\sin(\theta)}{P} \frac{d}{d\theta} \left(\sin(\theta) \frac{dP}{d\theta} \right) + \frac{1}{Q} \frac{d^2 Q}{d\phi^2} = 0 \quad (120)$$

Isolate the last term

$$\frac{r^2 \sin^2(\theta)}{R} \frac{d^2 R}{dr^2} + \frac{\sin(\theta)}{P} \frac{d}{d\theta} \left(\sin(\theta) \frac{dP}{d\theta} \right) = -\frac{1}{Q} \frac{d^2 Q}{d\phi^2} \quad (121)$$

Equate it to a constant (m)

$$\frac{1}{Q(\phi)} \frac{d^2 Q(\phi)}{d\phi^2} = -m^2 \quad (122)$$

Rearrange to get common differential equation solution

$$\frac{d^2 Q(\phi)}{d\phi^2} = -m^2 Q(\phi) \quad (123)$$

Common solutions if $m \neq 0$

$$Q(\phi) = A_m e^{im\phi} + B_m e^{-im\phi} \quad (124)$$

Common solutions if $m = 0$

$$Q(\phi) = A_m + B_m \phi \quad (125)$$

Back to Original equation 122, multiply $\sin^2(\theta)$

$$\frac{r^2}{R(r)} \frac{d^2 R(r)}{dr^2} + \frac{1}{\sin(\theta)P(\theta)} \frac{d}{d\theta} \left(\sin(\theta) \frac{dP(\theta)}{d\theta} \right) - \frac{m^2}{\sin^2(\theta)} = 0 \quad (126)$$

First and Last terms are independent so they can be set to constants

$$\frac{r^2}{R(r)} \frac{d^2 R(r)}{dr^2} - l(l+1) = 0 \quad (127)$$

Where

$$-l(l+1) = \frac{1}{\sin(\theta)P(\theta)} \frac{d}{d\theta} \left(\sin(\theta) \frac{dP(\theta)}{d\theta} \right) - \frac{m^2}{\sin^2(\theta)} \quad (128)$$

Put equation 127 into more intuitive forms

$$\frac{d^2 R(r)}{dr^2} = l(l+1) \frac{R(r)}{r^2} \quad (129)$$

Put equation 128 into more intuitive forms

$$\frac{d}{d\theta} \left(\sin(\theta) \frac{dP(\theta)}{d\theta} \right) + (l(l+1) - \frac{m^2}{\sin^2(\theta)}) \sin(\theta) P(\theta) = 0 \quad (130)$$

Reduce equation 130 to simpler form by substituting $x = \cos(\theta)$ and $\frac{d}{d\theta} = -\sqrt{1-x^2} \frac{d}{dx}$

$$\frac{d}{dx}((1-x^2)\frac{dP(X)}{dx}) + (l(l+1) - \frac{m^2}{1-x^2})P(x) = 0 \quad (131)$$

Equation 128 by substituting $R(r) = r^\alpha$

$$\alpha(\alpha-1)r^{\alpha-2} = (l+1)r^{\alpha-2} \quad (132)$$

$$\alpha^2\alpha - 1 - l(l+1) = 0 \quad (133)$$

$$\alpha = l + 1$$

$$\alpha = -l$$

General solution is

$$R(r) = A_l r^{l+1} + B_l r^{-1} \quad (134)$$

General Solution if the whole azimuthal sweep is included

$$\Phi(r, \theta, \phi) = \sum_m \sum_l (A_l r^l + B_l r^{-l-1})(A_m d^{im\phi} + B_m e^{-im\phi} P_l^m(\cos(\theta))) \quad (135)$$

Solving for $P(\theta)$ part we consider a specific case ($m = 0$) into equation 131

$$\frac{d}{dx}((1-x^2)\frac{dP(X)}{dx}) + l(l+1)P(x) = 0 \quad (136)$$

Guess, or "Try", power series solution

$$P(x) = \sum_{j=0}^{\infty} a_j x^{j+\alpha} \quad (137)$$

$$\sum_{j=0}^{\infty} (j+\alpha)(j+\alpha-1)a_j x^{j+\alpha-2} - (j+\alpha)(j+\alpha+1)a_j x^{j+\alpha} + l(l+1)a_j x^{j+\alpha} = 0 \quad (138)$$

Consolidate

$$\sum_{j=0}^{\infty} (j+\alpha)(j+\alpha-1)a_j x^{j+\alpha-2} - \sum_{j=0}^{\infty} ((j+\alpha)(j+\alpha+1) + l(l+1))a_j x^{j+\alpha} = 0 \quad (139)$$

Remove first two powers of x then combine

$$(\alpha)(\alpha-1)a_0 x^{\alpha-2} + (1+\alpha)(\alpha)a_1 x^{\alpha-2} - \sum_{j=0}^{\infty} ((j+2+\alpha)(j+\alpha+1)a_{j+2} - ((j+\alpha)(j+\alpha+1) - ((j+\alpha)(j+\alpha+1))a_j)x^{j+\alpha} \quad (140)$$

This must hold for all values of x so

$$(\alpha)(\alpha-1)a_0 = 0 \quad (141)$$

$$(1 + \alpha)(\alpha)a_1 = 0 \quad (142)$$

Substitute equation

$$(j + \alpha + 2)(j + \alpha + 1)a_{j+2} - ((j + \alpha)(j + \alpha + 1 - l(l + 1))) = 0 \quad (143)$$

The third equation is solved to yield the recurrence relation which gives all the remaining terms from the last one. Rerrange equation

$$a_{j+2} = \frac{(j + \alpha)(j + \alpha + 1) - l(l + 1)}{(j + \alpha + 2)(j + \alpha + 1)}a_j \quad (144)$$

$a_1 = 0$; therefore, $a_{odd} = 0$

$$P_l(x) = \sum_{j=0, even}^{\infty} a_j x^{j+\alpha} \quad (145)$$

Convergest at $x = 1$

$$\frac{(j_{max} + \alpha)(j_{max} + \alpha + 1) - l(l + 1)}{(j_{max} + \alpha + 2)(j_{max} + \alpha + 1)} = 0 \quad (146)$$

If $a = 0$

$$j_{max}(j_{max} + 1) - l(l + 1) = 0 \quad (147)$$

Therefore

$$j_{max} = l \quad (148)$$

Substitute equation

$$P_l(x) = \sum_{j=0, even}^l a_j x^j \quad (149)$$

Therefore

$$P_l(x) = \sum_{j=0}^{\frac{l}{2}} a_{2j} x^{2j} \quad (150)$$

Rewrite sums over all integers by doubling indices. If l is even

$$a_{j+2} = j \frac{(j + 1) - l(l + 1)}{(j + 2)(j + 1)}a_j \quad (151)$$

If $\alpha = 1$

$$(j_{max} + 2)(j_{max} + 1) - l(l + 1) = 0 \quad (152)$$

Therefore

$$j_{max} = l - 1 \quad (153)$$

Substitute equation

$$P_l(x) = \sum_{j=0, even}^{l-1} a_j x^{j+1} \quad (154)$$

Rewrite sums over all integers by doubling indices. If l is odd

$$a_{j+2} = \frac{(j+1)(j+2) - l(l+1)}{(j+3)(j+2)} a_j \quad (155)$$

Explicitly state low values of l

$$\begin{aligned} P_0(x) &= a_0 \\ P_1(x) &= a_0 x \\ P_2(x) &= a_0 + a_2 x^2 = a_0(1 - 3x^2) \text{ where } a_2 = -3a_0 \\ P_3(x) &= a_0 x + a_2 x^3 = a_0(x - \frac{5}{3}x^3) \text{ where } a_2 = -\frac{5}{3}a_0 \end{aligned}$$

a_0 is a scale factor and it is conventional to set $P_l(1) = 1$. This conditions allows us to set the scale factor. Additionally, x is a placeholder for the more functional $\cos(\theta)$

$$\begin{aligned} P_0(\cos(\theta)) &= 1 \\ P_1(\cos(\theta)) &= \cos(\theta) \\ P_2(\cos(\theta)) &= \frac{1}{2}(-1 + 3\cos^2(\theta)) \\ P_3(\cos(\theta)) &= \frac{1}{2}(-3x + \frac{5}{3}\cos^3(\theta)) \end{aligned}$$

Legendre polynomials have the following mathematical properties

$$\text{Odd or even symmetry about origin } P_l(-x) = (-1)^l P_l(x)$$

$$\text{Rodrigues' Formula } P_l(x) = \frac{1}{2^l l!} \frac{d^l}{dx^l} (x^2 - 1)^l$$

$$\text{Recurrence relation } P_{l+1}(x) = \frac{2l+1}{l+1} P_l(x) - \frac{l}{(l+1)} P_{l-1}(x)$$

$$\text{Rodrigues' Formula } P_l(x) = \frac{1}{2^l l!} \frac{d}{dx} (P_{l+1}(x) - P_{l-1}(x))$$

$$\text{Orthogonality condition } \int_{-1}^1 P_{l'}(x) P_l(x) dx = \frac{2}{2l+1} \delta_{l'l}$$

$$\text{Orthogonality condition (Polar angle)} \int_0^\pi P_{l'}(\cos(\theta)) P_l(\cos(\theta)) \sin(\theta) d\theta = \frac{2}{2l+1} \delta_{l'l}$$

The Legendre Polynomials form a complete set of orthogonal functions on the interval $(-1, 1)$, so any function $f(x)$ can be expanded in terms of Legendre Polynomials

$$f(x) = \sum_{l=0}^{\infty} A_l P_l(x) \quad (156)$$

Where

$$A_l = \frac{2l+1}{2} \int_{-1}^1 f(x) P_l(x) dx \quad (157)$$

The general solution to the Laplace Equation in spherical coordinates for the case $m = 0$ (where the origin and non-origin are considered)

$$\Phi(r, \theta, \phi) = \sum_{l=0}^{\infty} (A_l r^l + B_l r^{-l-1}) P_l(\cos(\theta)) \quad (158)$$